

PHYSICS XII

10 Year Questions Distribution

According to the Chapters.

This Distribution is carried out According to
NEW BOOK OF PHYSICS.

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QUESTIONS		YEARS
Unit:15	MOLECULAR THEORY OF GASES	
Section B		
xi) Calculate the root mean square speed of hydrogen gas molecule at 0°C and 1 atm pressure. Assuming hydrogen gas is to be an ideal gas and having density $8.99 \times 10^{-2} \text{ kg/m}^3$. Given $1 \text{ atm} = 1.01 \times 10^5 \text{ N/m}^2$		2025 P.E. PM
(vi) Calculate the temperature at which the root mean square speed of nitrogen molecule is 3300 ms^{-1} .		2023
(iii) Calculate the temperature at which the root mean square speed of hydrogen molecules is 3300 m/s . Give your answer in degree Celsius ($3.32 \times 10^{-27} \text{ kg}$)		2017
2(i) On the basis KMT of gases, show that $\frac{1}{2} m \bar{v}^2 = \frac{3}{2} KT$.		2015
Section C		
3. a) State the basic assumptions of kinetic molecular theory of gases. Derive the equation for pressure of an ideal gas, where $P = \frac{1}{3} \rho \bar{v}^2$ and ρ represents the density of the gas.		2025 P.E. PM
3. a) Derive a relation for the pressure of an ideal gas in terms of its density and mean square speed.		2025 S. G.
3. a. Derive an expression for the pressure of an ideal gas in terms of its density and mean square speed.		2024 S. G.
5. a) Derive pressure formula of an ideal gas using kinetic molecular theory.		2022
3 (a).Derive an expression for the pressure of an ideal gas in terms of its density and mean square speed.		2017

Unit:16	FIRST LAW OF THERMODYNAMICS	
Section B		
ix) Derive its mathematical expression $C_p - C_v = R$.	2025	
vii) A system undergoes a thermodynamic process where the pressure is kept constant at $15kPa$. If the volume increases from $0.05m^3$ to $0.08m^3$ and the change in internal energy is $150 J$. Calculate the heat added to the system.	P.E. PM	
xi) How much heat will be required to raise the temperature of $1 kg$ of lead from $30^\circ C$ to $105^\circ C$? (specific heat of lead = $0.128 J/kg^\circ C$)		
x) Define specific heat and molar specific heat and derive the relation between them.	2025	
xiii) How much heat is required to raise the temperature of $1 kg$ lead from $25^\circ C$ to $100^\circ C$? (Specific heat of lead = $128.1 J.kg^{-1}K^{-1}$)	S. G.	
4. b) Define heat capacity and specific heat and prove that $C_p = R + C_v$.	2024	
(vi) A system absorbs 1147 joules of heat, loses 233 joules of heat by conduction to the surroundings and delivers 614 joules of work. Calculate the change in the internal energy of the system.	P.E. PM	
	2018	
Section C		
4. b) State the first law of thermodynamics and explain following on its basis: i) Isobaric Process ii) Isochoric Process iii) Isothermal Process iv) Adiabatic Process	2023	
3 (b). Show that the difference of molar specific heat capacity at constant pressure and molar specific heat capacity at constant volume for an ideal gas is equal to the universal gas constant.	2018	
5 (a). State the first law of thermodynamics and explain on its basis. (i) isobaric process (ii) Isothermal process.	2015	

Unit:17	SECOND LAW OF THERMODYNAMICS	
Section B		
ii) A heat engine performs $200J$ of work in each cycle and has efficiency of 20 percent for each cycle of operation. How much heat is absorbed and how much heat is expelled?	2024	P.E. PM
vii) The difference of temperature between hot and cold body is $80^\circ C$. If the engine is 35% efficient, find the temperatures of hot and cold bodies?	2024	S. G.
2. i) A Carnot engine works between $800^\circ C$ and $400^\circ C$, if source temperature is increased by $50^\circ C$ or sink temperature is decreased by $50^\circ C$, then which will cause greater efficiency.	2022	
(vi) A system absorbs 1147 joules of heat, loses 233 joules of heat by conduction to the surroundings and delivers 614 joules of work. Calculate the change in the internal energy of the system.	2018	
(xi) Write the two statements of the second law of thermodynamics and prove their equivalence.		
2(i) The high temperature reservoir of a Carnot engine is at $200^\circ C$ and has an efficiency of 35% to increase the efficiency to 45% by how many degrees should the temperature of cold reservoir be decreased if the temperature of the high temperature reservoir remains constant?	2016	
(vi) A heat engine performs $200 J$ of work in each cycle and has an efficiency of 30 percent. For each cycle of operation, (a) how much heat is absorbed? (b) how much heat is expelled?	2015	
Section C		
4. b) What is Carnot Engine? Describe the process of a Carnot cycle and derive mathematical expression for its efficiency.	2025	P.E. PM
4. a) What is Carnot engine? Describe its construction and working & derive the expression for its efficiency.	2025	S. G.
3. a) What is Carnot engine? Derive an expression for the efficiency of Carnot engine.	2021	
3. (a) What is Carnot engine? Give its construction and working. Also derive mathematical expressions for its efficiency.	2019	
5 (b). Describe Carnot cycle and derive an expression for the efficiency of Carnot heat engine.	2016	

Unit:18	MAGNETIC FIELDS
Section B	
vi) A galvanometer has a resistance of 50Ω and it gives full scale deflection at 4 mA . A voltmeter is made using this galvanometer by connecting a $5k\Omega$ resistor in series. Calculate the maximum voltage that can be measured using this voltmeter.	2025 P.E. PM
xii) Define magnetic flux. Under what condition the magnetic flux will be maximum?	
iv) A moving coil galvanometer has resistance of 50Ω and it gives full scale deflection at 4 mA current. A voltmeter is made using this galvanometer with $5k\Omega$ resistance. Find the maximum voltage that can be measured with this voltmeter.	2025 S. G
ix) A solenoid with 400 turns has a cross section area of 4 cm^2 . A current of 2 A passing through it produces $B = 0.5\text{ tesla}$. How large an emf is produced in it? If the current is turned off is 0.1 second . What is its self-inductance?	2024 P.E. PM
xiii) A voltmeter measuring upto 150 volts has a total resistance of 15000 ohms . What additional resistance must be connected to it to increase its range to 450 volts ?	2024 S. G.
(iv) Derive an expression for the force experienced by a current carrying conductor in a uniform magnetic field.	2023
(xiv) An iron core solenoid with 400 turns has a cross-section area of 4.0 cm^2 . A current of 2.0 ampere passing through it produces $B = 0.2\text{ weber/m}^2$. What emf is produced in it if the current is turned off in 0.1 sec ? What is its self-inductance?	
ii) "What amount of current is required to produce a magnetic field of 37.2 Tesla at a distance of 2 cm from a long straight current carrying wire? ($\mu_0 = 4\pi \times 10^{-7}\text{ Weber/Am}$)	2022
ix) A 400 volt voltmeter has a total resistance of $40,000\Omega$. What additional series resistance must be connected to it to increase the range to 750 volts ?	
iii) A galvanometer has a resistance of 50Ω and it deflects full scale when a current of 500 micro ampere flows in it. How can it be converted into ammeter of range 15 ampere and voltmeter of range 300 volts ?	2021
2. (xiii) How can a galvanometer be converted into a voltmeter? Derive the relevant mathematical expression?	2019
(xiii) How can a galvanometer be converted into an ammeter?	2018

Derive the relevant expression.	
(xiii) Derive an expression for the force experienced by a current carrying conductor in a uniform magnetic field.	2017
(v) An iron core solenoid with 600 turns has a cross section area of 2.0 cm^2 . A current of 4.0 ampere passing through it produces $B=0.4\text{ weber/m}^2$. What is its self-inductance?	2016
(x) How is a galvanometer converted into voltmeter? Give the related mathematical expression.	
(xi) A proton of charge $1.6 \times 10^{-19}\text{ C}$ and mass $1.67 \times 10^{-27}\text{ kg}$ is accelerated by a potential difference of $6 \times 10^5\text{ volts}$. Then it enters perpendicularly into a magnetic field of intensity 0.5 Tesla . Find the radius of the circular path of the proton.	
(iii) Derive an expression for the force experienced by a current-carrying conductor in a uniform magnetic field.	2015
(viii) A galvanometer, whose resistance is 60 ohms , deflects full scale for a potential diff. of 100 millivolts across its terminals. What shunt resistance must be connected to convert it into an ammeter of 5 ampere range?	
Section C	
3. b) State Ampere's law. Derive an expression for the magnetic field of induction 'B' due to current passing through a long Solenoid OR Toroid.	2025 P.E. PM
3. b) State Ampere's law. Using Ampere's law derive the expression for the magnetic field inside a current carrying solenoid OR Toroid.	2025 S. G.
4. a) With the help of diagram describe the construction and working of moving coil galvanometer. Show that the current flowing through it is directly proportional to deflection.	2024 P.E. PM
5. a) State Ampere's Law. Using Ampere's law derive an expression for the magnetic field of induction inside a current carrying Solenoid OR toroid.	
4. b. Explain the functions of various parts of a moving coil galvanometer and prove that the amount of current flowing is directly proportional to the angle of twist of the suspension strip.	2024 S. G.
5. a. State Ampere's Law. Derive the expression for magnetic field of induction inside a current carrying solenoid OR toroid.	
3. a) State Ampere's law. Apply this law to find the magnetic field of induction inside a solenoid.	2022
4. a) Describe the construction and working of a moving coil	

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Galvanometer. Show that the deflection produced in galvanometer is directly proportional to the current.	
4 b) What is galvanometer? Prove that current passing through the coil of galvanometer is directly proportional to the angle of deflection of the coil.	2021
3. (b) State Ampere's Law. Derive the expression for magnetic field for induction B inside a current-carrying solenoid.	2019
4. (a) Describe the construction and working of moving coil galvanometer. Also show that the deflection produced in the coil is proportional to the current passing through it.	
4 (a). State Ampere's law. Use it to derive the relation for the magnetic field of induction B at any point inside a current-carrying toroid.	2017
5 (b). Explain the functions of the various parts of a moving coil galvanometer and prove that the amount of current flowing is directly proportional to the angle of twist of the suspension.	
4 (b). Describe the construction and working of moving coil galvanometer with the help of a neat diagram. Show that the current (i) is directly proportional to deflection (θ).	2015
5 (b). State Ampere's law. Using it, derive an expression for the magnetic field of induction inside a long current-carrying solenoid.	
[OR 5(b)] Describe the method for determining the ratio of charge to mass ($\frac{e}{m}$) of an electron. Derive the relevant mathematical expression.	

Unit:19 ELECTROMAGNETIC INDUCTION

Section B

viii) What is the name of the EMF induced in a straight conductor moving with uniform velocity in a magnetic field?	2025 P.E. PM
ii) A conductor of length 0.4m moves with a velocity of 5m/s perpendicular to a magnetic field of 0.3T. Calculate the motional e.m.f induced in the conductor.	2025 S. G.
xiv) What is meant by Eddy current? How Eddy currents are produced?	
2. i) Define mutual inductance and prove that $N_2\phi_2 = M I_1$	2024 P.E. PM
v) An alternating current generator operates at 79Hz. The area of the coil is 500 cm ² . Calculate the number of turns in the coil	2024 S. G.

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when a magnetic field of induction 0.06 tesla produces a maximum potential difference of 149 volts?	
2. (iii) A coil of 400 turns in AC Generator having an area of 0.1 m ² is rotating in a magnetic field of 50 T. in order to generate a maximum voltage of 220 volts, how fast is the coil to be rotated? Express your answer in revolutions/second.	2019
(x) An alternating current generator operate at 79Hz. The area of the coil is 500cm ² . Calculate the number of turns in the coil when a magnetic field induction 0.06 tesla produces a maximum potential difference of 149volts.	2017
(ix) An e.m.f. of 45 millivolts is induced in a coil of 500 turns. When the current in a neighboring coil changes from 15 amps to 4 amps in 0.2 seconds,	2015
(a) What is the mutual inductance of the coils?	
(b) What is the rate of change of flux in the second coil?	
Section C	
5. a) Define is AC generator. Describe its principle and working. Derive the expression for the alternating e.m.f.	2025 S. G.
5. b) What is Transformer? On what principle it works? With the help of diagram give its construction and working. Derive expressions for its efficiency and turns ratio.	2024 P.E. PM
5. a. State Ampere's Law. Derive the expression for magnetic field of induction inside a current carrying solenoid OR toroid.	2024 S. G.
3. b) What is transformer? On what principle does it work? Derive mathematical relationship between emf and number of turns in the coils.	2021
4 (a). What is a Transformer? Write its types with the help of a clear diagram, give its construction and working and derive the relevant expression.	2016
5 (a). State Faraday's laws of Electromagnetic induction. Explain Mutual induction and derive an expression of mutual inductance.	

Unit:20 AC CIRCUITS

Section B

v) In an RLC circuit, the resistance (R) is 50 ohms, the inductance (L) is 0.1 H and the capacitance (C) is 50 μ F. calculate its resonance frequency.	2025 P.E. PM
viii) In an RLC series circuit R is 50 Ω , inductance L is 0.1H	2025

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capacitance C is $50\mu F$. Calculate the resonance frequency.	S. G.
Section C	

Unit:21	PHYSICS OF SOLIDS
Section B	
Section C	

Unit:22	SOLID STATE ELECTRONICS
Section B	
vi) What are electron-hole pair in a semiconductor and how they are produced?	2025 S. G.
v) Define rectification. Name different types of rectifier. With the help of circuit diagram give working of any one rectifier.	2024 P.E. PM
xii) What is p-n Junction? Explain the formation of potential barrier in p-n Junction.	2024 S. G.
(vii) What is meant by Rectification? How does a semiconductor diode work as a half wave rectifier?	2023
v) What is semi conductor diode? Describe the working of full wave rectifier using semiconductor diode with the help of a circuit diagram.	2022
iv) With the help of diagram give the normal working of npn transistor.	2021
(v) What is meant by rectification? How does a semiconductor diode work as a half wave rectifier?	2018
(xiv) What is transistor? Why is it so called? Show diagrammatically the better connection to pnp transistor and npn transistor, for its normal working.	2017
(xiii) Describe the function of a PN-Junction as half wave rectifier.	2016
Section C	

Unit:23	DIGITAL ELECTRONICS
Section B	
iv) Draw the circuit symbol of OR and AND gates and write Truth Tables for two inputs.	2025 P.E. PM

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vii) Draw the truth table for OR gate with three inputs for it.	2025 S. G.
Section C	

Unit:24	RELATIVITY
Section B	
xiii) Prove that the speed of a moving rod at which its relative length is equal to half of its proper length is $\frac{\sqrt{3}c}{2}$.	2025 P.Z. PM
iii) Write the two postulates of special theory of relativity. Discuss its any two consequences.	2025 S. G.
xii) At what speed is a particle moving if the mass is equal to three times of its rest mass?	
(xii) Find the speed at which mass of particle will be doubled.	2023
xii) Who will be the velocity and momentum of particle whose rest mass is m_0 and whose K.E. (Kinetic Energy) is equal to its rest mass energy.	2022
2. (xv) What will be the velocity and momentum of a particle whose rest mass is m_0 and kinetic energy is equal to twice of its rest mass energy	2019
(iii) Write the principle of relativity and the two postulates of special theory of relativity.	2018
(iv) What will be the relativistic velocity of a particle whose kinetic energy is twice of its rest mass energy?	2015
Section C	
5. a) Write the postulates of special theory of relativity. Explain any two of its results.	2025 S. G.
3. b. What are inertial and non-inertial frames of reference? Write the postulates of special theory of relativity and give its consequences.	2024 S. G.
3. c) State the basic postulates of the special theory of relativity. Explain the four important results arrived at by Einstein on the basis of this theory.	2021
3 (b). What are inertial and non-inertial frames of references? Also give the consequences of the special theory of relativity.	2017

Unit:25	QUANTUM PHYSICS
Section B	
ii) State the laws governing the black body radiations. Also write their mathematical expressions.	2025 P.E. PM
iii) Calculate the wavelength of X-rays. If energy of a photon emitted by the X-rays machine is $1.9878 \times 10^{-13} \text{ J}$.	2024 P.E. PM
iv) Sodium light of wavelength $5.893 \times 10^{-7} \text{ m}$ falls on a photocell. A negative stopping potential of 0.30 volt is needed to stop the electrons from reaching the collector. Find the work function of the material of the plate. Give answer in joule as well as in electron volt. ($c = 3 \times 10^8 \text{ m/s}$, $h = 6.634 \times 10^{-34} \text{ J.s}$, $e = 1.6 \times 10^{-19} \text{ C}$)	2024 S. G.
viii) What is meant by the terms Pair Production and Annihilation of matter? Also write their equations.	2023
(iii) X-rays of wavelength $2.64 \times 10^{-10} \text{ m}$ are used in Compton scattering process. Find the fractional change in wavelength for a scattering angle of 120° .	2022
iii) In photo electric experiment, a stopping potential of 1.25 volt is needed. Find the maximum kinetic energy and maximum speed of photo electron. $e = 1.6 \times 10^{-19} \text{ C}$; $m_e = 9.1 \times 10^{-31} \text{ kg}$	2021
vi) Sodium surface is shined with light of wavelength $3 \times 10^{-7} \text{ m}$. If the work function of Na is 2.46eV, Find the K.E of the photoelectrons and cut off wavelength $\lambda_c h = 6.63 \times 10^{-34} \text{ J.s}$, $c = 3 \times 10^8 \text{ m/s}$, $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$.	2019
2. (xiv) What is a perfect black body? What are Max Plank's assumptions to explain black body radiations? Also write Plank's law of black body radiation.	2017
(vi) X-rays of wavelength $3.64 \times 10^{-10} \text{ m}$ are used in Compton scattering process. Find the fractional change in wavelength for a scattering angle of 120° (Give $h = 6.63 \times 10^{-34} \text{ J.s}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$, $c = 3 \times 10^8 \text{ m/s}$)	
(xiv) A photon of wavelength $0.004 \text{ \AA}$ in the vicinity of a heavy nucleus produces an electron-positron pair. Find the kinetic energy of each particle of the pair in MeV, If the kinetic energy of positron is twice that of electron. ($m_e c^2 = 8.19 \times 10^{-14} \text{ joules}$, $h = 6.63 \times 10^{-34} \text{ J.s}$ and $c = 3 \times 10^8 \text{ m/s}$)	

(vi) Sodium surface is shown with light of wavelength $3 \times 10^{-7} \text{ m}$ Find the kinetic energy of the emitted photo electron and the cut-off wavelength of Sodium. Work function of sodium is 2.46 eV.	2016
Section C	
4. a) What is photoelectric effect? Define the following terminologies *work function *stopping potential *threshold frequency and derive Einstein's photoelectric equation.	2025 P.E. PM
4. b) What are X-rays? How they are produced? Describe the two types of X-ray spectra.	2025 S. G.
5. b) What is photoelectric effect? Derive Einstein's photoelectric equation on the basic of quantum theory in terms of frequency and wavelength.	2024 P.E. PM
3. a) What is Compton effect? Derive an expression for the Compton shift in wavelength.	2023
3. a) What Photoelectric Effect? Explain its important results. Derive Einstein's photoelectric equation.	2022
3. b) Explain Compton Effect and obtain an expression for the increase in the wavelength of the scatter photon.	2019
4. (b) What is photoelectric effect? What is meant by the terms: (i) Threshold frequency (ii) Work function (iii) Saturation current (iv) Stopping potential Derive Einstein's photoelectric equation in terms of frequency as well as wavelength.	2018
5 (a). Explain Compton's effect. Derive an expression of shift in wavelength of scattered photon.	2016
3 (b). What is Compton effect? Derive an expression for the Compton shift in the wavelength.	2015
3 (b). What is Photoelectric effect? Explain its important results. Also derive Einstein's photo-electric equation.	

Unit:26	ATOMIC PHYSICS
Section B	
xiv) What is meant by LASER? Define metastable state OR population inversion.	2025 P.E. PM
v) How He - Ne Laser works?	2025 S. G.

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xii) How much energy is needed to ionize a hydrogen atom originally in the ground state? ($h = 6.63 \times 10^{-34} \text{ J-s}$, $c = 3 \times 10^8 \text{ m/s}$, $R_H = 1.097 \times 10^7 \text{ m}^{-1}$)	2024 P.E. PM
xiv) What is LASER? What is meant by terms Meta stable state, Population inversion and Optical pumping?	2024 S. G.
2. i) Find the shortest and longest wavelength of emitted photons in hydrogen spectra in Paschen series.	2023
x) Hydrogen atom in ground state is excited by absorbing a photon of 12.15eV. Find the quantum number of this excited state.	2022
v) Light of wavelength 486.3nm is emitted by a hydrogen atom in Balmer series. What transition of hydrogen atom is responsible for this radiation? ($R_H = 1.097 \times 10^7 \text{ m}^{-1}$)	2021
2. (vi) What is the wavelength of 3 rd spectral line of Paschen Series in hydrogen atom? ($R_H = 1.097 \times 10^7 \text{ m}^{-1}$)	2019
(viii) A photon of what minimum energy is required to excite a hydrogen atom from $n=1$ to $n=3$? ($R_H = 1.097 \times 10^7 \text{ m}^{-1}$)	2018
(viii) How much energy is needed to ionize a hydrogen atom originally in ground state? (Give $h = 6.63 \times 10^{-34} \text{ J-s}$, $c = 3 \times 10^8 \text{ m/s}$ $R_H = 1.097 \times 10^7 \text{ m}^{-1}$)	2017
(xv) What is meant by the term Laser, metastable state and population inversion?	2016
(vii) Find the shortest and longest wavelength of emitted photons in Hydrogen spectra in Pfund series.	2016
(xi) Find the shortest wavelength of photon emitted in the Balmer series and determine its energy in eV. ($R_H = 1.097 \times 10^7 \text{ m}^{-1}$)	2015
Section C	
5. b. Give the postulates of Bohr's atomic theory. Derive expression for radius of n^{th} orbit of hydrogen atom.	2024 S. G.
5. a) Derive the expression for the following when $r_n = n^2 \left(\frac{4\pi\epsilon_0 h^2}{me^2} \right)$	2023
i) the energy of an electron in the n^{th} orbit of hydrogen atom. ii) The wavelength of photons emitted in the hydrogen spectrum.	
4. b) Derive an expression for total energy of the electron in the	2022

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n^{th} orbit of a hydrogen atom.	
4 c) State the basic postulates of Bohr's theory of atomic structure. Derive an expression for the radius of the n^{th} orbit of a hydrogen atom.	2021
5. (b) Give the postulates of Bohr's atomic theory. Derive expression for the (i) Radius of n^{th} orbit of Hydrogen atom (ii) Total energy of electron in n^{th} orbit of Hydrogen	2019
4 (b). State the basic postulates of Bohr's atomic theory of Hydrogen atom. Derive the expression for the radius of the n^{th} orbit of Hydrogen atom. Hence, show that the ratio of radius of the third orbit to that of the second orbit is 2.25	2018
4 (b). Using Bohr's atomic theory. Derive expressions for the following: (i) The energy of an electron in the n^{th} orbit of hydrogen atom. Given: $r_n = \frac{h^2 n^2}{Kme^2}$ (ii) The wavelength of photons emitted in the hydrogen spectrum.	2016
4 (a). State the basic postulates of Bohr's theory of atomic structure. Derive the expression for the radius of n^{th} orbit of a Hydrogen atom.	2015

Unit:27

NUCLEAR PHYSICS

Section B

2. i) The $^{32}_{15}\text{P}$ isotopes of phosphorus has a half-life of 14.28 days. Find its decay constant. x) State law of radioactive decay and write its mathematical expression.	2025 P.E. PM
2. i) State law of radioactive decay also give its formula. What is meant by activity "A" and relative activity? ix) Find the binding energy of ^7_3Li ? Given: *Mass of $^7_3\text{Li} = 7.0160\text{u}$ *Mass of proton = 1.0073u *Mass of Neutron = 1.0087u	2025 S. G.
vi) Define fission reaction. What is the difference between controlled and uncontrolled fission reactions? xiii) Find the binding energy of $^{126}_{52}\text{Te}$ ($m_p = 1.0078\text{u}$, $m_n = 1.0086\text{u}$, $m_{^{126}_{52}\text{Te}} = 125.9033\text{u}$ and $1\text{u} = 931.5\text{MeV}$)	2024 P.E. PM

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x) Write the basic equations representing the change in a parent nucleus after alpha, positive beta β^+ , negative beta β^- and gamma emission.	2024 S. O.
xi) Find the Q-value of the following reaction: ${}_{94}\text{Pu}^{239} \rightarrow {}_2\text{He}^4 + {}_{92}\text{U}^{235} + Q$, if the mass of ${}_{94}\text{Pu}^{239} = 239.0522\text{u}$, ${}_{92}\text{U}^{235} = 235.0439\text{u}$, ${}_2\text{He}^4 = 4.0026\text{u}$	
(v) The half-life of ${}_{104}\text{Po}^{210}$ is 140 days. By what percent does its activity will decrease per week?	2023
xi) If the number of atoms per gram of ${}_{88}\text{Ra}^{226}$ is 2.666×10^{21} and it decays with the half life of 1622 year. Find decay constant and activity of the sample.	2022
vii) It is more difficult to start a fusion reaction than the fission reaction why?	2021
2. (ii) If the number of atoms per gram of ${}_{88}\text{Ra}^{226}$ is 2.666×10^{21} and it decays with the half-life of 1622 years, find the decay constant and activity of the sample. (1 Year = 3.15×10^7)	2019
2. (xi) Describe the radioactive decay law. Give the relevant mathematical expression.	
(iv) Find the Q-Value of the reaction ${}_{94}\text{Pu}^{239} \rightarrow {}_2\text{He}^4 + {}_{92}\text{U}^{22} + Q$ The isotopic mass of Plutonium = 239.0522 u The isotopic mass of Uranium = 235.0439 u The isotopic mass of alpha particle = 4.0026 u	2018
(vii) The half life of ${}_{104}\text{Po}^{210}$ is 140 days. By what percent will its activity decrease per hour?	2017
(xi) A deuteron of mass $3.3431 \times 10^{-27}\text{kg}$ is formed when a proton of mass $1.6724 \times 10^{-27}\text{kg}$ and neutron of mass $1.6748 \times 10^{-27}\text{kg}$ combine. Calculate the mass defect and binding energy in MeV.	
(vi) Sodium surface is shown with light of wavelength 3×10^{21} and it decays with a half life of 1622 years. Find the activity and decay constant of the sample.	2016
(viii) The number of atoms per gram of ${}_{88}\text{Ra}^{226}$ is 2.666×10^{21} and it decays with a half life of 1622 years. Find the activity and decay constant of the sample.	
(xiv) Find the binding energy and the packing fraction in MeV of ${}_{52}\text{Te}^{126}$ given that $m_p = 1.0078\text{u}$, $m_n = 1.0086\text{u}$, $m_{\text{Te}} = 125.9033\text{u}$ and $1\text{u} = 931.5\text{MeV}$	
(v) Find the Binding energy and Packing fraction (B.E. per nucleon) of ${}_{52}\text{Te}^{126}$. Given that $m_p = 1.0078\text{u}$, $m_n =$	2015

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1.0086u, $m_{\text{Te}} = 125.9033\text{u}$ $1\text{u} = 931.5\text{MeV}$	
Section C	
5. b) What is a mass spectrograph? Explain its construction, principle and working. Derive its relevant mathematical expression.	2025 P.E. PM
4. a) Define Radioactivity. Explain the law of radioactive decay. Write the equations of the change in the parent nuclei α , β and γ decay.	2023
Unit:28 PARTICLE PHYSICS	
Section B	
(viii) Give the construction and working of Geiger Counter.	2023
iv) Write a note on Wilson Cloud Chamber.	2022
ix) Give the construction and working of Wilson Cloud chamber.	2021
2. (v) Describe the construction and working of Wilson cloud chamber.	2019
(ix) Name a device for making the path of ionizing particles and rays visible and describe its working.	2017
(xiii) Give construction & working of Wilson cloud chamber.	2015
Section C	
5.b) What is Wilson Cloud Chamber OR Geiger Counter? Explain construction and working of any one.	2024 P.E. PM
4 (b). Give the construction and working of Geiger counter.	2018
[OR 5 (b)] Give construction and working of Geiger Muller Counter.	2016